Swiss Reality Check

Open benchmark + measurable RAG fix for Apertus 1.5 · Hack Apertus Online Hackathon · Track 1B Swiss Voices

Hackathon	Hack Apertus — Online Phase (opening stage of multi-month series)
Track / Challenge	Track 1 — Apertus Readiness · **1B: Swiss Voices** — Localising Apertus to Swiss Languages, Knowledge and Values
Problem statement	Evaluate how well Apertus handles Swiss reality: transcribe local dialects, test domain-specific Swiss knowledge, and assess whether responses respect Swiss cultural norms and values.
Team	Swiss Reality Check · solo · **Bhuwan B** (bhuwanb23)
Models under test	Designed for both `swiss-ai/Apertus-v1.5-8B` and `swiss-ai/Apertus-v1.5-70B` (Apache-2.0, EPFL / ETH Zurich / CSCS). Live scores: **8B only**. 70B: not run (8 GB VRAM).
Deadline / links	Submit by 16 Oct 2026, 12:00 CEST · GitHub github.com/bhuwanb23/apertus · Dataset huggingface.co/datasets/bhuwanb23/apertus · Partners: CSCS, Phoeniqs · Grand Finals May 2027, St. Gallen

0. Challenge mapping (Track 1B)

Track 1B asks three evaluations of Swiss reality. This project maps them 1:1 onto measurable pillars and one contamination-safe improvement:

- **Domain-specific Swiss knowledge** → 120 source-verified MCQs (politics, law, health, geography, services, history); distractors = real DE/EU answers.
- **Swiss cultural norms & values** → 30 scenarios; rubric scores *framing / institutional competence*, not political side (+ multilingual pairs).
- **Local dialects** → 30 Alemannic (Swiss German) items from als.wikipedia.org; refusal rate is a headline metric. Text modality this round (Apertus 1.5 also supports audio; audio track not used here).
- **Improvement (Readiness)** → BM25 RAG over harvested Swiss sources, gated so the system prompt cannot leak answer keys.

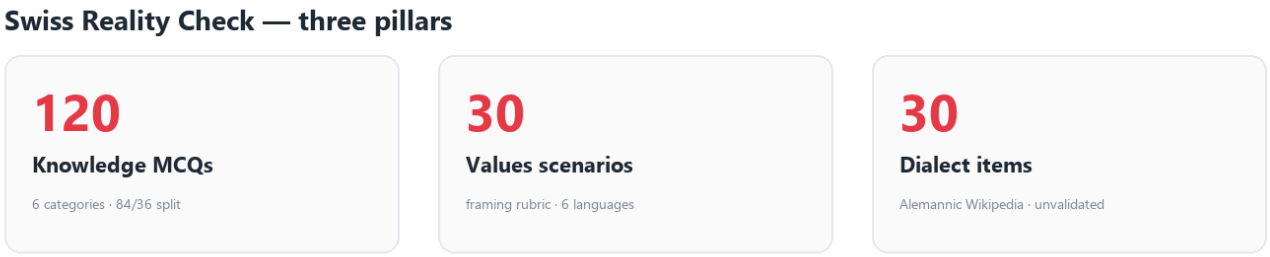


Figure A. Benchmark scale (180 core items). Split knowledge 84/36 sealed.

Headline (sealed test, 8B): 55.6% → 86.1% with BM25 RAG (+30.6 pp, McNemar 11↑/0↓, p=0.001). 70B: not evaluated.

1. Models under evaluation

Apertus is Switzerland's sovereign open LLM (EPFL, ETH Zurich, CSCS), trained across 1,800+ languages with fully open training data. v1.5 adds multimodal inputs, thinking mode, and 262k context. Hack Apertus expects work with **both** 8B and 70B; this submission documents the dual-model design and reports live numbers only where the GPU could host the weights.

Property	Apertus-v1.5-8B	Apertus-v1.5-70B
HF id	swiss-ai/Apertus-v1.5-8B	swiss-ai/Apertus-v1.5-70B
Family	Multilingual · multimodal (image/audio/text→text) · open weights+data	Same family; larger scale
Context	up to 262,144 tokens	up to 262,144 tokens
Pretraining	~19T tokens (incl. +4T multimodal mix on 1.0)	~17T tokens (incl. +2T multimodal mix on 1.0)
Architecture	Decoder-only Transformer · xIELU · AdEMAMix · bf16	Same architecture class
This project	**Measured** — text extract Q4_K_M via llama.cpp HTTP, T=0, seed 42	**Not measured** — out of scope on 8 GB VRAM; no imputed scores

Table 1. Model cards: huggingface.co/swiss-ai/Apertus-v1.5-8B and …/Apertus-v1.5-70B (License Apache-2.0). Serving for this project: community text-only Q4_K_M GGUF + llama.cpp HTTP · temperature 0.0 · seed 42 · is_mock=false on every artefact.

2. Method — RAG without answer-key contamination

Corpus: 100 harvested official / Alemannic pages → 189 non-dialect chunks. Retriever: pure-Python BM25, `top_k=3`. A plan-proposed fact-list system prompt would answer **25/84 (30%)** of development items from the prompt alone. Shipped behavioural prompt: **0/84** — hard-gated before every run (`fix/prompts_check.py`). Scorers for test = same functions as development (no reimplemented parsers). Seal: SHA-256 of knowledge_test.json + append-only access log.

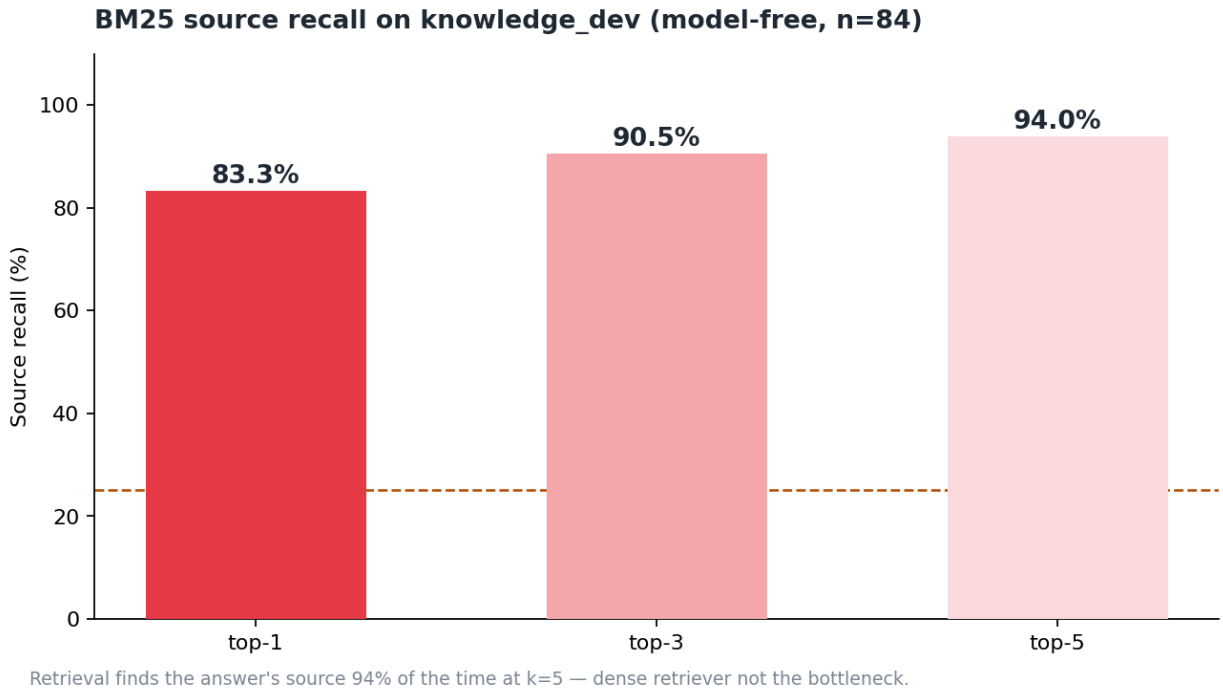


Figure B. Model-free source recall on knowledge_dev (n=84): 83.3 / 90.5 / 94.0% @ k=1/3/5. Dense retrieval was not the bottleneck.

3. Main result — Swiss knowledge (held-out test)

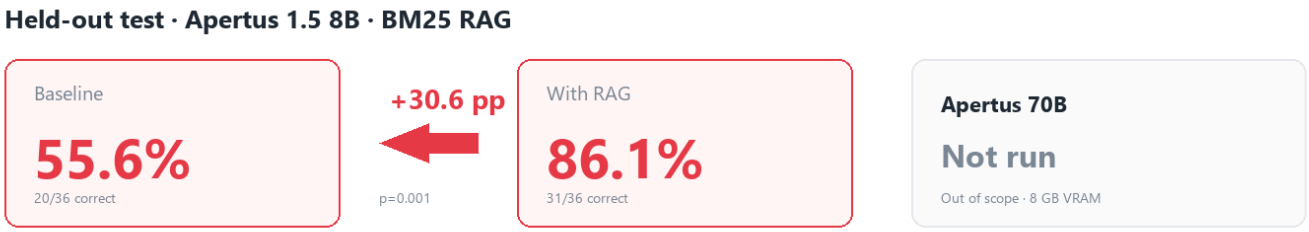


Figure C. Paired before/after on sealed n=36. 70B column is hardware scope, not silence.

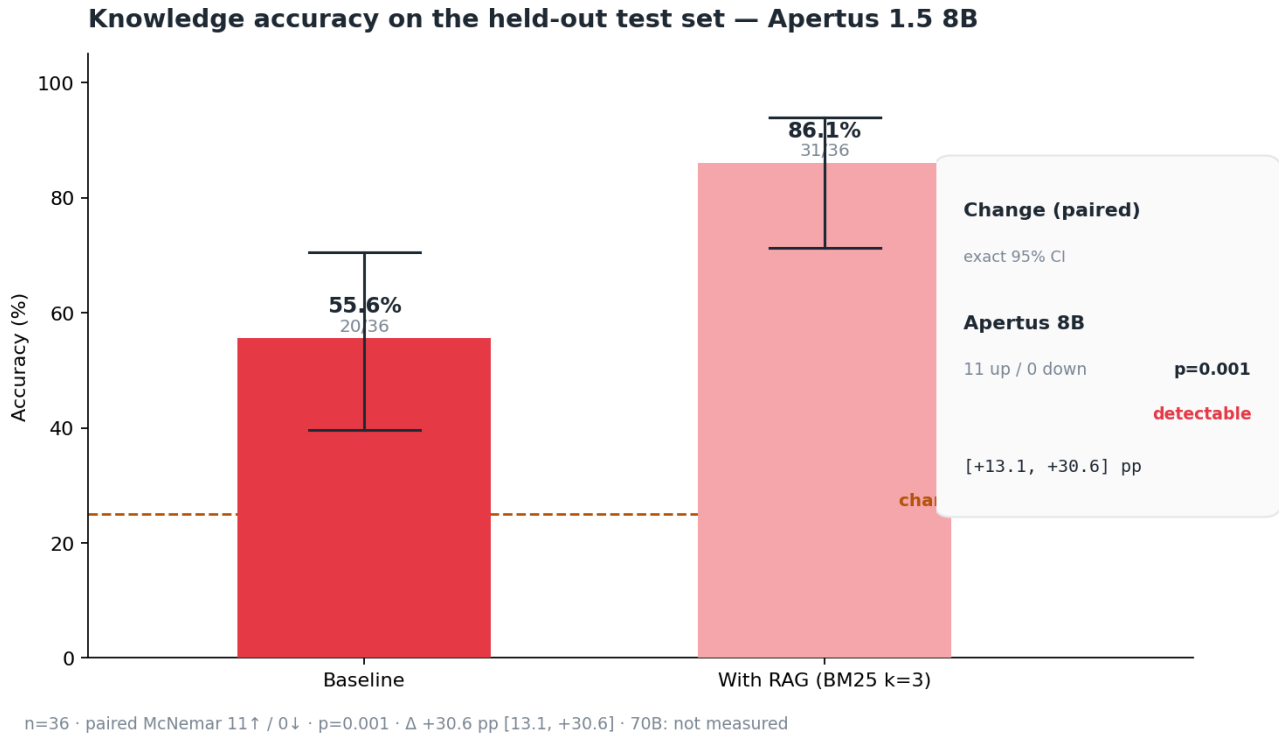


Figure 1. Accuracy + 95% Wilson intervals. Chance = 25%. Δ +30.6 pp, exact 95% CI [+13.1, +30.6]; McNemar p=0.001.

Arm	8B accuracy	95% Wilson	70B
Baseline (no retrieval)	55.6% (20/36)	[39.6, 70.5]	Not run
Improved (BM25 RAG k=3)	86.1% (31/36)	[71.3, 93.9]	Not run
Paired delta	+30.6 pp · p=0.001	[+13.1, +30.6]	—
Dev transfer (n=84)	48.8% → 83.3% (+34.5 pp)	paired McNemar	Not run

Power note: at n=36 a paired change needs ≥6 improvements with 0 regressions (+16.7 pp) to be detectable. This run cleared that floor (11/0).

4. Category detail · transfer · contamination

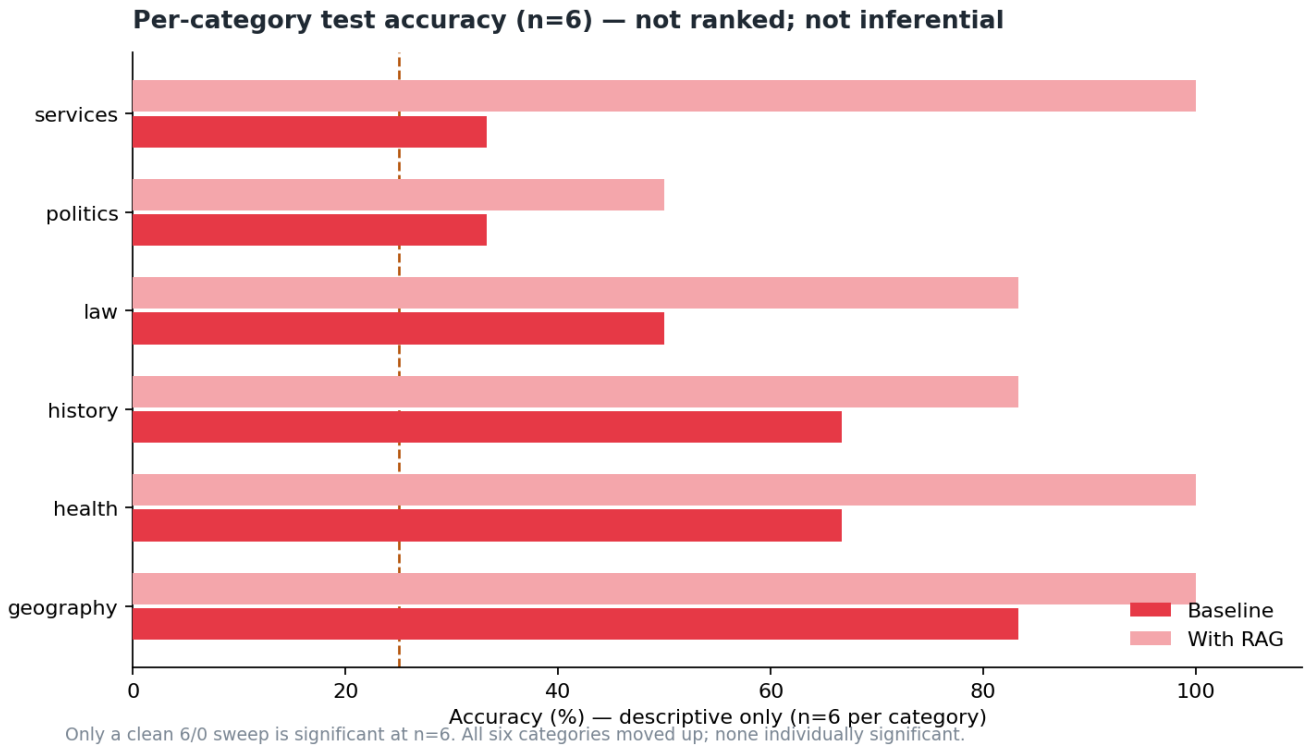


Figure 2. Per-category test accuracy (n=6). Descriptive only — only a clean 6/0 sweep is significant at this n. All six categories moved up; none individually significant. Not ranked (selection on the measurement set).

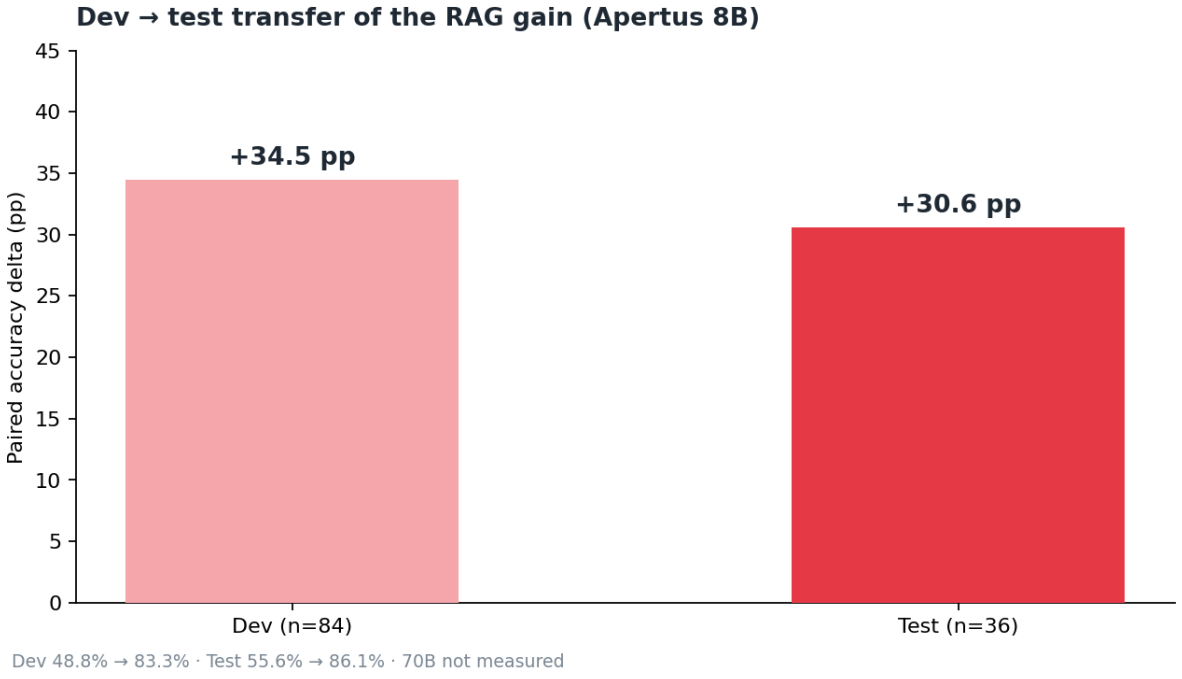


Figure 3. Dev→test transfer of the RAG gain. Survival of the effect on a sealed hold-out is the scientifically load-bearing claim for Track 1 Readiness.

Integrity disclosure: the sealed set was opened more than once due to a since-fixed `--resume` flag bug. Decoding was greedy (T=0, seed 42); nothing was tuned between opens. Reported figures are from the completed arms (55.6% / 86.1%), not a mean. Full write-up: eval/README.md Disclosures.

5. Values, dialect & dual-model scoreboard

Evaluation scope — honest dual-model scoreboard

Metric	8B (measured)	70B
Knowledge baseline	**55.6% (20/36)**	Not run
Knowledge + RAG	**86.1% (31/36)**	Not run
Paired delta	**+30.6 pp · p=0.001**	—
Values mean / 2.0	**1.096 (self-judged)**	Not run
Dialect refusal	**50.0% (15/30)**	Not run

Figure D. Honest 8B | 70B scoreboard — 70B cells are out of scope, never imputed from 8B.

5.1 Values & norms (Track 1B — cultural norms)

30 scenarios × multilingual renderings → **73** judged records. Mean **1.096 / 2.0**; distribution {0: 33, 2: 40}. Rubric scores epistemic framing (correct Swiss institutional frame, acknowledged disagreement, no DE/EU mis-attribution), not which political side the model takes. Judge = same 8B weights (**self-judged**, disclosed). Human blind audit (15 items, B5) **not completed** — no agreement rate claimed.

5.2 Dialect (Track 1B — local dialects)

30 items: 12 T1 translation, 8 T2 / 5 T3 / 5 T4 MCQ. Source sentences verbatim from Alemannic Wikipedia (CC BY-SA 4.0). Live 8B: refusal **50.0%** (15/30); MCQ **50.0%**; translation mean **0.0** (all T1 refused). All `validated_by: null`. Refusal is reported separately from accuracy — a model that declines Swiss German is a Track 1B finding, not a parse error.

5.3 Why 70B is blank (but still in scope of the design)

Hack Apertus provides 8B and 70B with CSCS / Phoeniqs compute. This solo setup ran on consumer 8 GB VRAM. Metadata records `scope.dropped: ["70b"]`. The harness already accepts a 70B HTTP/OpenAI backend; when hosted, the same sealed runner can fill Table 1 without changing scorers. Inventing 70B percentages would violate submission integrity.

6. Limitations

1. **n=36** — detection floor +16.7 pp (0 regressions). Per-category n=6 is descriptive.
2. **Single measured scale** — no live 8B-vs-70B comparison (hardware).
3. **Self-judged values** — no human agreement rate yet.
4. **Dialect unvalidated** — native-speaker packet ready, not signed off.
5. **Seal openings > 1** — disclosed; greedy decode; figures from completed arms.
6. **Text-only dialect** — Apertus 1.5 audio path unused this round.
7. **Temporal** — Swiss law / institutions as of October 2026.

7. Reproducibility & distribution

- Stats git: `436ba260830af773434675282a2cc897f15d13c9`
- Seed 42 · temperature 0.0 · prompt version 1.0
- Sealed test SHA-256: `df819e8c1582902a…`
- Code: `https://github.com/bhuwanb23/apertus`
- Dataset: `https://huggingface.co/datasets/bhuwanb23/apertus`
- Models: `https://huggingface.co/swiss-ai/Apertus-v1.5-8B` · `…/Apertus-v1.5-70B`
- License (project): MIT · Datasheet: DATASHEET.md (Gebru et al. 2021)
- Rebuild: `python scripts/prepare_submission_assets.py` · `python scripts/build_submission_pdf.py` · `python scripts/upload_hf_dataset.py`

Swiss Reality Check answers Track 1B with a sealed, source-grounded measurement and a contamination-safe RAG fix that transfers from development to hold-out on Apertus 8B. 70B remains an open measurement for CSCS-scale follow-up — not a fabricated claim. Everything ships open for reuse under MIT.

Team: Swiss Reality Check (Bhuwan B) · Hack Apertus Online · Track 1B Swiss Voices · 16 October 2026